

Factor-Augmenting Technology and the Measurement of Markups: Evidence from Indian Manufacturing, 2012–2022*

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Abstract

Production-approach markup estimates rest on an assumption so standard it is rarely stated: productivity differences across plants must be *Hicks-neutral*. If technology instead augments labour — plausible wherever mechanisation and managerial quality vary as much as they do in Indian manufacturing — standard estimators read technology heterogeneity as market power. We implement the factor-augmenting production-function estimator of [Demirer \(2025\)](#) — to our knowledge its first independent implementation — on 286,931 plant-year observations from India’s Annual Survey of Industries (2012–2022), estimating industry technologies with labour-augmenting heterogeneity by nested GMM with distribution-regression control functions, in 207 industry $\times$ rolling three-year windows covering all manufacturing divisions. The implementation is validated four ways: an analytic unit test; a Monte Carlo on a theory-consistent data-generating process with known technology (which also maps the estimator’s failure modes, including the functional-dependence degeneracy its genericity condition excludes); a full-fidelity re-estimation on real data; and firm-cluster bootstrap inference re-running the entire pipeline in every resample. The findings: the pooled median markup falls from 1.240 (translog) and 1.222 (Cobb–Douglas) to **1.118** under factor-augmenting technology — Hicks-neutral methods overstate the median Indian markup by roughly ten per cent — and the share of plants pricing above marginal cost falls from 94 to 79 per cent.

*Department of Humanities and Social Sciences, Indian Institute of Technology Roorkee. This paper is the fourth of a series based on a newly constructed plant panel from the Annual Survey of Industries; the companion papers measure markups and markdowns under Hicks-neutral technology (working paper 1), misallocation (working paper 2), and their macroeconomic determinants (working paper 3). ASI unit records are used under licence from the Ministry of Statistics and Programme Implementation and cannot be redistributed; our estimator code, Monte Carlo suite, and all industry-level results are available from the authors. All errors are ours.

The bias anatomy matches the original paper almost exactly: Cobb–Douglas overstates the flexible-input elasticity by 12.3 per cent (Demirer: about 11) and both Hicks-neutral estimators understate the labour elasticity by 29–36 per cent — so markdown-based measures of labour-market power estimated under Hicks-neutrality are *lower* bounds. The revision is statistically decisive (the translog–FA gap exceeds two bootstrap standard errors in 18 of 23 industries), concentrated exactly where technology heterogeneity is most plausible, and flattens the decade’s markup trend. Product-market power in India is even more modest than measured; labour-market power is larger. The binding margin is monopsony, not markups.

JEL codes: D24, C14, L11, L60, O14.

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Contents

1	Introduction	6
2	Related literature	9
3	The measurement problem, stated formally	10
3.1	What the production approach measures	10
3.2	What Hicks-neutrality assumes, and what it costs	11
3.3	The direction and size of the bias, made precise	11
4	The model and its identification	13
4.1	Technology: factor-augmenting heterogeneity with strong homotheticity . .	13
4.2	The two identities that deliver markups	14
4.3	Two unobservables, two control variables	15
4.4	Why one control function is not enough	15
4.5	Estimating equations and the combined objective	16
4.6	Functional forms and normalisations	17
5	Implementation	18
5.1	Step 1: the first control variable	18
5.2	Step 2: distribution regression for the second control variable	18
5.3	Step 3: design matrices	19
5.4	Step 4: nested optimisation	19
5.5	Documented deviations from the original settings	19
5.6	Computational footprint	20
6	Validation	20
6.1	Layer 1: analytic unit test	20
6.2	Layer 2: Monte Carlo on a theory-consistent DGP	20
6.3	Layer 3: full-fidelity re-estimation on real data	21
6.4	Layer 4: full-pipeline bootstrap inference	22
7	Data	22
8	Results	23
8.1	Elasticities: the bias anatomy	23
8.2	Markups: level, dispersion, rank	23
8.3	Reading the industry evidence	26
8.4	Inference	27
8.5	Dispersion: how much measured markup spread is technology?	29
8.6	The decade trend	29

8.7	What the correction means, economically	31
8.8	Placing the results against the original five-country study	32
9	Robustness	32
10	Limitations and extensions	33
11	Implications	34
12	Conclusion	35
A	Derivations	37
A.1	The elasticity-ratio identity (8)	37
A.2	Absorption of ω^L into the aggregator	37
B	The algorithm, step by step	38
C	Monte Carlo detail	38
D	Supplementary tables	39
D.1	Window log summary	39
D.2	Reproducibility	39

List of Tables

1	Findings at a glance (matched sample of 254,672 plant-years unless noted).	8
2	Full-fidelity check, 2017 window: production settings (grid 40, degree-2 nuisance) versus original settings (grid 300, degree-3), six industries.	21
3	Mean output elasticities: factor-augmenting versus Hicks-neutral, matched sample ($N = 254,672$). Percentage gaps relative to FA. Demirer (2025) reports a flexible-input overstatement of about +11 per cent across his five-country panel.	23
4	Markups under the three technologies, matched sample.	24
5	Median markups by industry: FA versus translog and Cobb–Douglas, matched sample. Gap = translog – FA.	24
6	Bootstrap inference, 2017 window ($B = 100$ firm-cluster resamples per industry, full pipeline re-estimated in each; all resamples converged). Gap = translog – FA median markup (matched sample); $t = \text{gap} / \text{s.e.}$	27
7	Median markup by year on the matched sample (window midpoints 2013–2021).	30

8	The markdown correction implied by the elasticity biases. Since $\psi = (\theta^L/\theta^M)(\alpha^M/\alpha^L)$ at common shares, the FA-versus-Hicks-neutral markdown ratio equals the ratio of θ^L/θ^M across technologies. Mean elasticities from Table 3; illustrative aggregate arithmetic (not a plant-level re-estimation).	35
9	Estimated aggregator coefficients and objective minima by industry: medians across the nine rolling windows. N = total midpoint plant-years entering results.	40

List of Figures

1	The Hicks-neutrality markup overstatement by industry: translog minus FA median markup, matched sample 2013–2021.	25
2	The markup revision measured against bootstrap uncertainty: (translog – FA median markup) / bootstrap standard error, by industry. Dashed lines at ± 2	28
3	The decade’s markup trend under three technology assumptions.	30

1 Introduction

The production approach to markup measurement — recovering the markup as the ratio of an output elasticity to a revenue share (Hall, 1988; De Loecker and Warzynski, 2012) — has become the workhorse of the market-power literature, in India as elsewhere. Its appeal is real: no demand system, no conduct assumption, no price data beyond revenues. Its central maintained assumption, however, is rarely stated, because it hides inside the production function itself: productivity differences across plants must be *Hicks-neutral*, shifting the marginal products of all inputs equiproportionally, so that a single scalar ω multiplies the whole technology. Under that assumption, two plants with different materials-to-labour ratios must differ in *prices or wedges*, never in technology — and the estimator, forced to rationalise input-mix heterogeneity somehow, loads it into the output elasticities and from there into measured markups.

The assumption is least defensible exactly where markup measurement is most consequential. In a manufacturing sector like India’s — where hand-loom and power-loom textile plants coexist within four-digit industries, where mechanisation, automation, and managerial quality vary across plants by an order of magnitude — the natural alternative is *labour-augmenting* heterogeneity: technology that multiplies effective labour, $\Omega^L L$, so that a better-managed or more mechanised plant behaves as if it had more workers, not more of everything. Demirer (2025) formalises the consequences. If true technology is factor-augmenting and the econometrician imposes Hicks-neutrality, the estimated flexible-input elasticity absorbs the technology-driven component of input intensity and is biased *upward*; the labour and capital elasticities are crowded *downward*; and since the markup is elasticity over share, measured markups inherit the biases — in level, in dispersion, and in trend. Demirer also supplies the constructive part: a nonparametric identification argument and estimator that recover the technology, and hence markups, *robustly* to labour-augmenting heterogeneity, at the cost of a genuinely harder estimation problem — two unobservables instead of one, control functions built from conditional distributions rather than simple inversions, and a nested optimisation in place of a regression. Applying it to plant data from five countries, he finds markups materially lower and less variable than Hicks-neutral methods report.

This paper implements that estimator on the modern Indian plant panel — the cleaned ASI factory panel 2012–2022 built for this series, an order of magnitude broader than the ≥ 100 -employee 1998–2014 Indian sample in the original paper — and asks the applied question that motivates the exercise: *how much of measured Indian market power survives factor-augmenting technology?* The question matters twice over here. The companion papers’ central empirical contrast — modest product-market power against pervasive labour-market power — was estimated entirely under Hicks-neutrality. If that assumption inflates markups, the contrast sharpens on one side; and because the markdown literature

divides the *labour* elasticity by the labour share, an understated labour elasticity understates monopsony one-for-one — so the same correction that deflates markups *inflates* measured labour-market power. The direction of every revision is an empirical question this paper answers.

Because the same plants carry the series’ Hicks-neutral estimates — an ACF Cobb–Douglas benchmark and the De Loecker–Warzynski translog used throughout — the comparison is exact: same plants, same years, same deflators, same revenue shares; only the technology assumption changes. Any difference in measured market power is attributable to that assumption alone. This is, to our knowledge, the first independent implementation of the estimator (no public replication package exists; we implement directly from the published algorithm and release our code), and the largest sample to which it has been applied.

Four questions organise the paper. **Q1 (identification):** how does one recover plant-specific technology when *two* unobservables — Hicks-neutral and labour-augmenting productivity — drive input choices, and what does the identification cost in assumptions? Section 4 develops the answer equation by equation. **Q2 (implementation):** can the estimator be implemented faithfully, and validated, outside its author’s code base? Sections 5–6 document every step, every deviation, and four layers of validation — an analytic unit test, a Monte Carlo with known technology (whose failed first drafts turned out to be as informative as its final pass), a full-fidelity re-estimation, and a full-pipeline bootstrap. **Q3 (measurement):** how much do Indian markups, elasticities, and their decade trend change when Hicks-neutrality is dropped? Section 8 quantifies. **Q4 (consequences):** what do the answers do to the Indian market-power and misallocation debate? Section 11 traces the implications through the companion papers’ results.

The findings, in brief. *Levels:* the pooled median factor-augmenting (FA) markup is 1.118 against 1.240 (translog) and 1.222 (Cobb–Douglas) on the identical 254,672 matched plant-years — a ten per cent overstatement of the level of Indian product-market power under Hicks-neutrality, appearing in 21 of 23 divisions, largest exactly where cross-plant technology heterogeneity is most plausible (tobacco +0.74, apparel +0.37, pharmaceuticals +0.27, electronics +0.26) and vanishing in process industries where technology is embodied in capital (basic metals −0.01, printing −0.02). *Elasticities:* mean output elasticities under FA are 0.80 (materials), 0.16 (labour), 0.04 (capital) with returns to scale 1.00; Cobb–Douglas puts 0.90 on materials (+12.3 per cent, against the +11 Demirer reports) and 0.11 on labour (−29 per cent) — the bias anatomy of the original paper, reproduced on new data by independent code. *Dispersion and rank:* the FA distribution is tighter (s.d. 0.30 vs 0.54 Cobb–Douglas) — part of measured markup dispersion is technology dispersion in disguise — while cross-plant rankings survive (Spearman 0.72 with Cobb–Douglas). *Trend:* the FA series is nearly flat (1.10–1.13 over nine years) where the translog rises to a 2015 peak and falls by 0.08: Hicks-neutral methods overstate the

decade’s markup *movement*, not just its level. *Inference*: with firm-cluster bootstrap standard errors from 100 full-pipeline resamples per industry (all 2,300 converged), the downward revision exceeds two standard errors in 18 of 23 industries and three in 16.

The paper contributes to three literatures. To the *measurement debate* on production-approach markups (Bond et al., 2021; Raval, 2023; De Loecker and Scott, 2024; Demirer, 2025; Rubens et al., 2025) it contributes the first independent replication of the factor-augmenting estimator, a validated public implementation, a Monte Carlo cartography of the estimator’s failure modes, and a five-country-to-India confirmation of the bias anatomy. To the *Indian market-power literature* (De Loecker et al., 2016) and this series it contributes the correction with the sharpest consequences: the thesis’s central asymmetry — small markups, large markdowns — is not an artefact of Hicks-neutrality but is *strengthened* by relaxing it, in both directions at once. To the small literature taking non-neutral technology seriously in firm data (Doraszelski and Jaumandreu, 2018; Zhang, 2019; Raval, 2023) it adds developing-country evidence at plant scale that the neutrality assumption is quantitatively first-order for exactly the objects — markups, markdowns, their dispersion — that the applied literature estimates.

Table 1 collects the headline numbers. Section 2 situates the paper. Section 3 states the measurement problem formally. Section 4 develops the model and its identification. Section 5 documents the implementation; Section 6 its validation. Section 7 describes the data. Section 8 presents results; Section 9 robustness; Section 11 implications; Section 12 concludes. Appendices derive the key identities (A), give the algorithm in full (B), document the Monte Carlo (C), and report supplementary tables (D).

Table 1: Findings at a glance (matched sample of 254,672 plant-years unless noted).

Object	Estimate
Estimation runs	207 industry×window cells; 23 divisions; 2013–2021 midpoints
Median markup: FA / translog / Cobb–Douglas	1.118 / 1.240 / 1.222
Share of plants with $\mu > 1$: FA / translog	0.787 / 0.942
Markup s.d.: FA / translog / CD	0.300 / 0.239 / 0.540
Materials-elasticity bias (CD vs FA)	+12.3% (Demirer: $\approx +11\%$)
Labour-elasticity bias (CD / TL vs FA)	−29.4% / −35.7%
Returns to scale (FA)	0.998
Spearman rank corr. (FA, CD) / (FA, TL)	0.721 / 0.234
Industries with TL > FA median markup	21 of 23
Bootstrap: gap > 2 s.e. / > 3 s.e.	18 / 16 of 23 industries
FA markup trend 2013–2021	1.109 $\rightarrow$ 1.132 $\rightarrow$ 1.104 (band of 0.03)
Monte Carlo recovery (truth $\theta^V = .85$, $\mu = 1$)	$\widehat{\theta^V} \approx 0.89\text{--}0.93$; $\widehat{\mu} \approx 1.03\text{--}1.09$

2 Related literature

The production approach and its critics. Hall (1988) showed that market power breaks the equality between output elasticities and revenue shares that perfect competition implies; De Loecker and Warzynski (2012) turned the wedge into a firm-level estimator, $\mu = \theta/\alpha$ for any statically flexible input, with the elasticity estimated by control-function methods (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Akerberg et al., 2015). The ensuing decade of applications — including the influential rise-of-market-power literature (De Loecker et al., 2020) — provoked an equally influential scrutiny. Bond et al. (2021) prove that with revenue (not quantity) data, the level of the markup is not identified without demand-side information — what is estimated are revenue elasticities — disciplining all level claims. Raval (2023) documents that markups computed from labour versus materials disagree systematically — ranked oppositely, trending differently — which rejects some part of the joint assumptions in most applications, and points to non-neutral technology as the leading suspect. Doraszelski and Jaumandreu (2018) had already shown, for Spanish firms, that labour-augmenting technical change is large and heterogeneous. Demirer (2025) closes the circle: he shows formally that under factor-augmenting technology the Hicks-neutral estimators are biased in signed, quantifiable directions; that the labour-versus-materials disagreement Raval documents is exactly what factor-augmenting technology predicts; and that an estimator built on strong homotheticity plus a two-control-function inversion recovers technology robustly. De Loecker and Scott (2024) triangulate production-side against demand-side markups; Rubens et al. (2025) push the non-neutrality agenda into labour-market power. Our contribution to this strand is replication and validation: an independent implementation, from the published algorithm alone, that reproduces the original bias anatomy on new data — the kind of evidence a measurement debate needs most and gets least.

The Raval disagreement, and how the FA model resolves it. It is worth dwelling on Raval (2023), because his diagnostic is the empirical hook on which the whole factor-augmenting agenda hangs. Raval computes the production-approach markup twice for the same plant — once from labour, θ^L/α^L , once from materials, θ^M/α^M — and observes that under the maintained Hicks-neutral model the two *must* be equal (both equal μ), yet in data they diverge systematically: they are ranked differently across plants and trend differently over time. Something in the joint assumptions is false. The FA model identifies the culprit precisely. Under factor-augmenting technology the labour-based ratio is not μ but $\mu \cdot \psi^{-1}$ -adjusted by the augmentation — the two input ratios differ by exactly the labour-augmenting wedge — so their disagreement is not a failure of the data but a *measurement* of ω^L . Equation (10) shows the FA estimator makes the labour- and materials-implied markups coincide by construction *because* it has already extracted the augmentation into the aggregator $\bar{h}$; what was an unexplained disagreement

in the Hicks-neutral world becomes the identifying variation in the FA world. Our results are, in this light, a constructive resolution of Raval’s puzzle on Indian data: the input-choice ambiguity he documents is exactly the technology heterogeneity the FA estimator recovers.

India. The Indian markup literature spans the liberalisation studies (De Loecker et al., 2016) and the newer plant-panel estimates of this series (papers 1–3), all Hicks-neutral. India is also one of Demirer’s five countries (ASI 1998–2014, plants with 100+ employees); our sample is the full ASI frame a decade forward — roughly an order of magnitude more plant-years — making the exercise both a replication (same country, different decade, independent code) and an extension (full size distribution, matched Hicks-neutral comparators on identical data, bootstrap inference, and the markdown implications the original paper does not pursue).

Non-neutral technology in firm data. Beyond Doraszelski and Jaumandreu (2018), Zhang (2019) estimates non-neutral technology and labour demand in Chinese data; Oberfield and Raval (2021) aggregate micro elasticities of substitution into macro technology, with factor-augmenting heterogeneity central to the mapping. The applied lesson common to this literature — and the one our results carry to India — is that treating technology differences as scalar is not a harmless normalisation; it is a substantive restriction that the data can, and here do, reject in economically meaningful magnitudes.

3 The measurement problem, stated formally

3.1 What the production approach measures

For a plant minimising cost over a statically flexible input X purchased at price p^X in a frictionless market, the first-order condition of the cost-minimisation problem equates the input’s marginal revenue product to its price at the shadow value of the output constraint, and rearranges to

$$\mu_{it} \equiv \frac{P_{it}}{MC_{it}} = \frac{\theta_{it}^X}{\alpha_{it}^X}, \quad \theta_{it}^X = \frac{\partial \ln Y_{it}}{\partial \ln X_{it}}, \quad \alpha_{it}^X = \frac{p_{it}^X X_{it}}{P_{it} Y_{it}}, \quad (1)$$

the output elasticity over the revenue share. The share is read from the accounts; the elasticity must be estimated — and *everything* in this paper concerns how that estimation depends on what one assumes about technology. Two facts about (1) frame the exercise. First, it holds for *any* flexible input, so a plant’s labour-implied and materials-implied markups should coincide; their systematic disagreement in applied work (Raval, 2023) indicts the maintained assumptions. Second, the elasticity in (1) is plant-and-time specific; an estimator that forces it to be common within an industry (Cobb–Douglas) or to vary only with observed input mix (translog under Hicks-neutrality) restricts where markup

heterogeneity is allowed to come from.

3.2 What Hicks-neutrality assumes, and what it costs

The standard empirical model writes, in logs,

$$y_{it} = f(k_{it}, l_{it}, m_{it}; \beta) + \omega_{it} + \varepsilon_{it}, \quad (2)$$

with one scalar unobservable ω entering *additively in logs* — equivalently, multiplying the whole production function in levels. This is Hicks-neutrality: a high- ω plant gets proportionally more output from every input bundle, so relative marginal products — and hence cost-minimising input *ratios* at common prices — are identical across plants with the same (k, l, m) . The assumption does real work in every step of the standard method: the scalar ω is what the materials-demand inversion of [Levinsohn and Petrin \(2003\)](#) inverts; its scalarity is what makes the control function one-dimensional; and its neutrality is what lets input ratios identify the technology parameters β .

Now suppose the truth is factor-augmenting: plants differ also in ω^L , which multiplies labour, $Y = F(K, e^{\omega^L} L, M) e^{\omega^H + \varepsilon}$. A high- ω^L plant gets more from its workers, so at common prices it optimally uses *less* materials per (raw) worker — input ratios now vary across plants for technological reasons. The Hicks-neutral estimator (2), which by construction attributes all systematic input-ratio variation to the shape of f , misreads that technology variation as curvature: the flexible-input elasticity is biased up, the labour elasticity down, and the capital elasticity down ([Demirer, 2025](#), Section 3). Through (1) the biases propagate one-for-one into markups (numerator too big) and markdowns (labour elasticity too small). And because ω^L is heterogeneous *across plants* and persistent *within* them, the contamination is not classical noise: it inflates cross-sectional markup dispersion and can manufacture markup *trends* out of technology diffusion. Every one of these predictions is testable by estimating both ways on the same data — which is what this paper does.

3.3 The direction and size of the bias, made precise

It is worth deriving *why* the flexible-input elasticity is overstated, because the argument shows the bias is not a numerical accident but a structural feature — and because it predicts exactly the cross-industry pattern the results display. Consider a stripped-down version of the two models with a constant-returns inner aggregator, and focus on the materials elasticity. Under the true factor-augmenting technology, a cost-minimising plant

chooses materials per raw worker according to

$$\tilde{m}_{it} = m_{it} - l_{it} = \underbrace{\kappa_t(p^M/w)}_{\text{prices}} + \underbrace{\phi \omega_{it}^L}_{\text{technology}}, \quad \phi = \frac{1}{1-\rho} > 0, \quad (3)$$

where ρ governs substitution in the inner aggregator: a plant with higher labour-augmenting productivity ω^L has *more* effective labour, so at common prices it uses more materials per raw worker (the sign of ϕ is positive whenever labour and materials are gross complements or mild substitutes, $\rho < 1$). Equation (3) is the crux: *materials intensity varies across plants partly for a technological reason*, ω^L , that the Hicks-neutral estimator does not model.

Now ask what a Hicks-neutral estimator does with this variation. That estimator regresses output on inputs treating the single residual ω as the only unobservable; because it has no ω^L in its model, the systematic covariation between materials intensity $\tilde{m}$ and output — which the FA model attributes to ω^L raising *both* $\tilde{m}$ (through (3)) and output (through effective labour) — is loaded onto the only channel available, the curvature of f in materials. Formally, letting $\widehat{\theta}^{HN}$ denote the Hicks-neutral estimate,

$$\mathbb{E}[\widehat{\theta}^{HN}] - \theta^M = \underbrace{\phi \frac{\partial y}{\partial \omega^L}}_{>0} \cdot \underbrace{\frac{\text{var}(\omega^L)}{\text{var}(\tilde{m})}}_{\text{share of } \tilde{m} \text{ variation from technology}} > 0, \quad (4)$$

a positive bias whose magnitude rises with (i) how much labour-augmenting productivity raises output, $\partial y / \partial \omega^L$, and (ii) how much of the observed dispersion in materials intensity is technological rather than price-driven, $\text{var}(\omega^L) / \text{var}(\tilde{m})$. Three consequences follow, each borne out in Section 8. First, the flexible-input elasticity is *always* overstated under Hicks-neutrality (the right side of (4) is signed), so the markup numerator is always too big — a one-directional bias, not noise. Second, the overstatement is *largest exactly where technology heterogeneity is largest* — high $\text{var}(\omega^L)$ — which is why the markup revisions in Section 8.2 are biggest in tobacco and apparel (hand-versus-machine coexistence) and smallest in continuous-process industries. Third, because the true elasticity vector sums to (roughly) constant returns, the materials overstatement must be paid for by *understating* labour and capital — the crowding-out that makes Hicks-neutral markdowns too small (Section 11). Equation (4) is heuristic — Demirer (2025) gives the rigorous statement without the constant-returns shortcut — but it captures the economics the full estimator makes operational and tells the reader in advance what pattern of results to expect and why.

4 The model and its identification

This section develops the estimator’s logic equation by equation: the technology and the dimension-reduction that strong homotheticity buys (4.1); the two identities that deliver markups (4.2); the two-unobservable inversion and its control functions (4.3); and the estimating equations and moment conditions (4.5).

4.1 Technology: factor-augmenting heterogeneity with strong homotheticity

Plant i in year t produces with

$$Y_{it} = F_t(K_{it}, H_t(e^{\omega_{it}^L} L_{it}, M_{it})) e^{\omega_{it}^H + \varepsilon_{it}}, \quad (5)$$

where K is capital, L labour, M materials; ω^L is *labour-augmenting* productivity (entering only through effective labour $e^{\omega^L} L$), ω^H is Hicks-neutral productivity, and ε is an ex-post shock unknown when inputs are chosen. The inner aggregator H_t combines the two flexible inputs into a *variable-input bundle*; *strong homotheticity* is the assumption that H_t is homogeneous of degree one (its shape does not depend on scale), while the outer function F_t is unrestricted in (K, H) . Three remarks unpack what (5) assumes and buys.

What it nests. Setting ω_{it}^L constant across plants recovers exactly the Hicks-neutral model (2) with a nested CES/Cobb–Douglas-like inner structure; so the FA model is a strict generalisation, and comparing estimates across the two is a specification test with economic content, not a horse race between non-nested models.

What homotheticity buys. Because H_t is homogeneous of degree one, $H_t(e^{\omega^L} L, M) = e^{\omega^L} L \cdot h_t(M/(e^{\omega^L} L))$ for some scalar function h_t — the bundle is effective labour times a function of the materials-to-*effective*-labour ratio. In logs, defining $\tilde{m}_{it} = m_{it} - l_{it}$ (log materials per raw worker),

$$\ln H_t = \omega_{it}^L + l_{it} + \ln h_t(e^{\tilde{m}_{it} - \omega_{it}^L}), \quad (6)$$

so the *only* way ω^L enters is through the combination $\tilde{m} - \omega^L$ and an additive level. This is the crucial dimension reduction: a two-dimensional technology heterogeneity problem collapses to tracking one composite inside a known structure.

How ω^L is then absorbed. Under cost minimisation, the plant’s optimal materials-per-effective-worker ratio depends on prices and ω^L alone; inverting the materials-demand rule gives $\omega_{it}^L = \bar{r}_t(\tilde{M}_{it})$ — the observed materials-to-labour ratio reveals labour-augmenting productivity (monotonically, within a year and industry, at common prices). Substituting this inversion into (6) makes $\ln H_t$ a function of l and $\tilde{m}$ *only*, and the empirical model

becomes

$$\boxed{y_{it} = f_t(k_{it}, v_{it}) + \omega_{it}^H + \varepsilon_{it}, \quad v_{it} = l_{it} + \ln \bar{h}_t(\tilde{m}_{it}),} \quad (7)$$

Demirer’s equation (5.1): output is an unknown function of capital and a *composite variable input* v , itself built from labour and an unknown scalar aggregator $\bar{h}_t$ of the materials-to-labour ratio. Everything plant-specific about labour-augmenting technology now lives inside $\bar{h}_t$ evaluated at the plant’s own $\tilde{m}$ — estimated, not assumed. What breaks (7): input prices that vary across plants *within* an industry-year (the inversion then confounds prices with ω^L); non-homothetic inner structure (the ratio no longer suffices); or measurement error in m or l that varies systematically with productivity. The first concern is why estimation runs within narrowly defined industry-windows; the third is diagnosed in the Monte Carlo (Section 6), where price noise violating strong homotheticity produces a recognisable failure signature.

4.2 The two identities that deliver markups

Both L and M are statically flexible, so each satisfies its version of the cost-minimisation condition behind (1). Taking the ratio of the two first-order conditions — Appendix A gives the two-line derivation — the markup and every unobservable cancel, leaving

$$\frac{\theta_{it}^M}{\theta_{it}^L} = \frac{\alpha_{it}^M}{\alpha_{it}^L} : \quad (8)$$

elasticity ratios equal revenue-share ratios, plant by plant, whatever the markup and whatever the technology parameters. Equation (8) is both a gift and a discipline. The gift: define the *bundle elasticity* $\theta^V := \theta^L + \theta^M = \partial f_t / \partial v$ (the derivative of (7) with respect to the composite input); then (8) splits it using observed shares,

$$\theta_{it}^L = \theta_{it}^V \frac{\alpha_{it}^L}{\alpha_{it}^V}, \quad \theta_{it}^M = \theta_{it}^V \frac{\alpha_{it}^M}{\alpha_{it}^V}, \quad \alpha_{it}^V := \alpha_{it}^L + \alpha_{it}^M, \quad (9)$$

so only the bundle elasticity needs estimating — the individual elasticities follow by arithmetic. And the markup, computed from either input, is

$$\boxed{\mu_{it} = \frac{\theta_{it}^V}{\alpha_{it}^V} = \frac{\theta_{it}^L}{\alpha_{it}^L} = \frac{\theta_{it}^M}{\alpha_{it}^M}.} \quad (10)$$

The discipline: (10) says the labour-implied and materials-implied markups coincide *by construction* — the estimator is immune to the input-choice ambiguity that Raval (2023) shows afflicts Hicks-neutral applications. Conversely, where Hicks-neutral estimates of θ^L/α^L and θ^M/α^M disagree in applied work, (8) says the disagreement itself is evidence of factor-augmenting technology: under the FA model the two ratios cannot differ for a

cost-minimising plant, so their measured gap is a specification diagnostic, not a nuisance.

4.3 Two unobservables, two control variables

Estimating (7) still faces simultaneity: v and k are chosen knowing ω^L and ω^H . The Hicks-neutral toolkit inverts one scalar unobservable from one input demand; here there are two unobservables, and the inversion must be two-dimensional. Demirer’s construction uses two *rank* variables:

$$u_{it}^1 = F_{\tilde{m},t}(\tilde{m}_{it}), \quad (11)$$

the within-window empirical CDF of the materials-to-labour ratio — by the monotone inversion $\omega^L = \bar{r}_t(\tilde{M})$, u^1 is exactly the plant’s *rank in the ω^L distribution*, a normalisation-free control for labour-augmenting productivity; and

$$u_{it}^2 = F_t(m_{it} | k_{it}, W_{it-1}, u_{it}^1), \quad W_{it-1} = (k_{it-1}, l_{it-1}, m_{it-1}), \quad (12)$$

the *conditional* CDF of materials given capital, the lagged input vector, and u^1 . The logic of (12) deserves its intuition. Conditional on where the plant stands in the ω^L distribution (u^1), on its capital, and on its history (W_{t-1}), the remaining variation in its materials purchase is driven by ω^H — so the plant’s conditional rank in materials is its rank in the *Hicks-neutral* productivity distribution. Together (u^1, u^2) control for both unobservables nonparametrically, the way a single inverted materials demand controls for one ω in Levinsohn and Petrin (2003). Identification then requires *genericity* (Demirer’s Proposition 5.1(iii)): the input-demand system must supply variation that is not an exact function of the conditioning variables — if input rules are deterministic, linear functions of few state variables, the control functions absorb all the variation that would identify f , and the problem degenerates. This condition sounds technical; Section 6 shows it binding concretely in simulation, and explains why real plant data — with idiosyncratic input variation from prices, frictions, and mistakes — sit comfortably inside the generic set.

4.4 Why one control function is not enough

It is instructive to see precisely where the standard toolkit breaks, because it locates what the FA estimator adds. The Levinsohn and Petrin (2003)/Akerberg et al. (2015) approach rests on a *scalar-unobservable* assumption: a single productivity ω drives both output and the proxy (materials) demand, so materials demand $m_t(k, l, \omega)$ is invertible in ω , and conditioning on (k, l, m) purges the unobservable. The inversion is one equation in one unknown — it works because there is exactly one thing to solve for. With factor-augmenting technology there are *two* unobservables, ω^H and ω^L , entering the plant’s decisions; one input demand cannot invert two unknowns. Materials demand now takes the form $m = m_t(k, l, \omega^H, \omega^L)$, and conditioning on (k, l, m) leaves a one-dimensional

locus of (ω^H, ω^L) pairs consistent with the data — the unobservable is not pinned down, and any Hicks-neutral inversion silently projects the two-dimensional truth onto a one-dimensional proxy, mislabelling the projection residual as technology curvature (the bias of Section 3.3).

The FA estimator restores identification by using *two* rank statistics rather than one level inversion. The first, $u^1 = F(\tilde{m})$, exploits that the materials-to-labour ratio is monotone in ω^L alone (equation (3)), so a plant’s *rank* in the $\tilde{m}$ distribution is its rank in the ω^L distribution — one unobservable located. The second, $u^2 = F(m \mid k, W_{-1}, u^1)$, then uses the residual variation in the *level* of materials, after conditioning on the ω^L -rank and the state, to locate the plant in the ω^H distribution — the second unobservable. Two controls for two unobservables: dimension matched to dimension. This is why the estimator needs the conditional distribution, not just a conditional mean — a mean would collapse the rank information the identification requires — and why it needs *both* the ratio $\tilde{m}$ (for ω^L) and the level m (for ω^H), objects a Hicks-neutral estimator treats as redundant. The cost is a genuinely harder computation, but the payoff is that technology heterogeneity is *estimated* rather than assumed away.

4.5 Estimating equations and the combined objective

Two estimating equations operationalise the model, mirroring the two-stage logic of [Akerberg et al. \(2015\)](#) but with the FA structure. The *first equation* projects output on the technology and the controls:

$$y_{it} = f(k_{it}, v_{it}; \mathbf{a}, \mathbf{b}) + c_2(W_{it-1}, u_{it}^1, u_{it}^2) + \varepsilon_{it}^{(1)}, \quad (13)$$

where c_2 — a flexible polynomial — absorbs $\mathbb{E}[\omega^H \mid W_{t-1}, u^1, u^2]$, so the error $\varepsilon^{(1)}$ contains only the ex-post shock and the projection residual. Equation (13) is Demirer’s (5.3); it identifies the technology’s *shape* given the aggregator parameters $\mathbf{a}$ inside v . The *second equation* imposes the Markov assumption on Hicks-neutral productivity, $\omega_{it}^H = g(\omega_{it-1}^H) + \xi_{it}$ with $\mathbb{E}[\xi_{it} \mid \mathcal{I}_{it-1}] = 0$: subtracting the estimated technology from output and projecting on the lagged state,

$$y_{it} - f(k_{it}, v_{it}; \mathbf{a}, \mathbf{b}) = c_3(W_{it-1}) + (\xi_{it} + \varepsilon_{it}^{(2)}), \quad (14)$$

Demirer’s (5.4), whose composite error — productivity innovation plus noise — must be orthogonal to anything in the plant’s lagged information set and to predetermined capital.

The estimator stacks both requirements into the *combined objective*, his (5.5):

$$J(\mathbf{a}) = \underbrace{\frac{1}{N} \sum_{it} \hat{\varepsilon}_{it}^{(1)2}}_{\text{fit of (13)}} + \underbrace{\sum_j \left[\frac{1}{N} \sum_{it} z_{j,it} (\hat{\xi}_{it} + \hat{\varepsilon}_{it}^{(2)}) \right]^2}_{\text{moments of (14)}}, \quad z_j \in \text{poly}(k_{it}, W_{it-1}). \quad (15)$$

Read the two pieces as division of labour. The least-squares part disciplines *fit*: given a candidate aggregator, the technology and controls must explain output. The moment part disciplines *exogeneity*: the implied productivity innovations must be unpredictable by current capital and every lagged input — the same timing logic as the Hicks-neutral literature (capital predetermined by time-to-build; lagged flexible inputs chosen before ξ realised; relevance from the persistence of productivity and input demands). What makes (15) genuinely different from a two-stage ACF procedure is that the aggregator parameters $\mathbf{a}$ enter *inside* the definition of v , so the objective cannot be concentrated into a linear problem globally — but *given* $\mathbf{a}$, everything else ($\mathbf{b}$, c_2 , c_3) enters linearly. That structure dictates the nested optimisation of the next section.

4.6 Functional forms and normalisations

The nonparametric objects are approximated exactly as in the original paper’s empirical implementation. The aggregator is a normalised cubic (his OA.31),

$$\ln \bar{h}(\tilde{m}) = a_2 \tilde{m} + a_3 \tilde{m}^2 + a_4 \tilde{m}^3, \quad a_1 = 0, \quad (16)$$

where the normalisation $a_1 = 0$ pins the level of v (only the *shape* of the aggregator is identified — a constant would be absorbed by f); and the technology is the ten-term cubic (his OA.32)

$$f(k, v) = b_1 k + b_2 k^2 + b_3 k^3 + b_4 v + b_5 v^2 + b_6 v^3 + b_7 k^2 v + b_8 k v^2 + b_9 k v \ (+b_0), \quad (17)$$

whose analytic derivatives give the plant-level elasticities,

$$\theta_{it}^V = b_4 + 2b_5 v_{it} + 3b_6 v_{it}^2 + b_7 k_{it}^2 + 2b_8 k_{it} v_{it} + b_9 k_{it}, \quad \theta_{it}^K = b_1 + 2b_2 k_{it} + 3b_3 k_{it}^2 + 2b_7 k_{it} v_{it} + b_8 v_{it}^2 + b_9 v_{it}, \quad (18)$$

evaluated at each plant’s own (k, v) — elasticities, and hence markups (10), are plant-specific through both the cubic curvature and the plant’s position in it. Estimation runs separately for every NIC-2 industry $\times$ rolling three-year window (midpoints 2013–2021), so all functions carry implicit industry-window subscripts, and results are read at each window’s midpoint year: the rolling structure lets technology drift across the decade without forcing a single f on eleven years of data.

5 Implementation

No public replication package exists for the original paper; the implementation was written directly from the published algorithm (Online Appendix E.1) in Python (NumPy/SciPy/scikit-learn), and this section documents it completely — Appendix B gives the step-by-step algorithm with tolerances. The unit of estimation is one industry-window; the pipeline below runs 207 times.

5.1 Step 1: the first control variable

u^1 is the empirical CDF (11) of $\tilde{m}$ within the window, computed by double-argsort ranking with ties broken stably and scaled to $(0, 1)$ by $(\text{rank})/(N + 1)$ — the standard positive-probability normalisation that keeps logits finite downstream.

5.2 Step 2: distribution regression for the second control variable

u^2 requires the *conditional* CDF (12), estimated by distribution regression (Foresi and Peracchi, 1995): over an equal-count grid of G quantile points $q_1 < \dots < q_G$ of m , fit a logit of the indicator $\mathbf{1}\{m_{it} \leq q_g\}$ on a polynomial of $(k_{it}, W_{it-1}, u_{it}^1)$, giving each plant a fitted probability $\hat{P}_g(x_{it})$ of lying below each grid point; then evaluate each plant’s own conditional CDF at its own m_{it} by linear interpolation between the two neighbouring grid points. Degenerate grid cells (indicator mean below 0.5 per cent or above 99.5) are assigned their unconditional mean rather than fitted, and the interpolated CDF is clipped to $[10^{-4}, 1 - 10^{-4}]$. Distribution regression is the right tool here because it delivers the conditional CDF *at each plant’s own outcome* — precisely the rank object the identification argument needs — while remaining a sequence of convex, well-understood logit fits rather than a multidimensional density estimation problem.

The estimand deserves a sentence of precision, since the whole identification rests on it. What (12) requires is $u_{it}^2 = \Pr(M_{it} \leq m_{it} \mid k_{it}, W_{it-1}, u_{it}^1)$ — the probability that a plant with this capital, history, and ω^L -rank draws a materials level at or below its *own* realised level. Distribution regression estimates the entire conditional CDF by running, at each threshold q_g , the regression $\Pr(M \leq q_g \mid X) = \Lambda(X'\beta_g)$ with Λ the logistic link and a *separate* coefficient vector β_g at every threshold — so the shape of the conditional distribution is free to change across the support, unlike a single-index or homoskedastic model. Evaluating the fitted CDF at the plant’s own m_{it} (by interpolation between the two bracketing thresholds) is the Foresi and Peracchi (1995) conditional-rank transform: under correct specification it is uniform conditional on X , which is exactly the property that makes it a valid control for the second unobservable. The two deviations from the original settings here — 40 thresholds rather than 500, degree-2 rather than degree-3 index

polynomials — coarsen the grid and the flexibility of each β_g , but leave the estimand and its interpretation untouched; the fidelity check (Section 6.3) confirms the coarsening does not move the economic outputs.

5.3 Step 3: design matrices

Given (u^1, u^2) , three fixed design blocks are assembled once per window: the control polynomial c_2 in (W_{t-1}, u^1, u^2) ; the Markov polynomial c_3 in W_{t-1} ; and the instrument block $\mathbf{z}$, a full degree-3 polynomial in $(k_t, k_{t-1}, l_{t-1}, m_{t-1})$, standardised column-wise so the moment penalties in (15) are scale-free. Year dummies enter both c_2 and c_3 , absorbing within-window common shocks (each window spans three years).

5.4 Step 4: nested optimisation

The estimation exploits the structure noted under (15): *given* the aggregator coefficients $\mathbf{a} = (a_2, a_3, a_4)$, the composite input $v = l + a_2\tilde{m} + a_3\tilde{m}^2 + a_4\tilde{m}^3$ is data, and both estimating equations are linear — (13) is one OLS of y on $[f\text{-terms}, c_2]$, and (14) is one OLS of $y - \hat{f}$ on c_3 . The objective $J(\mathbf{a})$ is therefore a three-dimensional *profile* function, minimised by Nelder–Mead from three starting points ($\mathbf{a}_0 \in \{(0.5, 0, 0), (0.9, 0, 0), (0.3, .05, 0)\}$; convergence tolerances 10^{-4} in the simplex and 10^{-7} in the objective; best of the three retained). Nelder–Mead is appropriate because the profile is low-dimensional, cheap (each evaluation is two least-squares solves), and not guaranteed smooth in finite samples; the multi-start guards against local minima, and in practice the three starts agree. Elasticities (18) are then evaluated at every plant in the window’s *midpoint* year, and markups follow from (10) with the plant’s own nominal shares.

5.5 Documented deviations from the original settings

Four deviations from the original paper’s settings, all imposed by computational budget and all validated in Section 6: (i) the distribution-regression grid uses $G = 40$ equal-count points against his 500; (ii) the distribution-regression and control polynomials (c_2 , c_3 , and the logit design) are degree 2 against his 3 — while the *economic* objects, the aggregator (16) and technology (17), retain the original cubic forms exactly; (iii) bootstrap inference uses $B = 100$ firm-cluster resamples per industry at the 2017 window against his 250; (iv) the Hicks-neutral comparators are this series’ ACF Cobb–Douglas and DLW translog rather than his Blundell–Bond system estimator — deliberately, since the comparison of interest here is with the estimates the Indian literature actually uses. The Monte Carlo shows (i)–(ii) are innocuous on synthetic data — full-fidelity settings (degree 3, 300-point grid) are indistinguishable from the fast settings on identical simulated samples — and the real-data fidelity check (Section 6.3) confirms it on six representative industries.

5.6 Computational footprint

The full production run — 207 industry-windows, window sizes from roughly a thousand to fifteen thousand plant-years — completes in 23,310 seconds (6.5 hours) single-threaded; the bootstrap (2,300 full-pipeline re-estimations) in 40,720 seconds. Every window converged; the outer minimiser’s objective J and the estimated (a_2, a_3, a_4) for all 207 windows are archived in the window log (per-industry summaries in Appendix Table 9).

6 Validation

An independent implementation of a new estimator owes the reader evidence that it implements what it claims. We supply four layers, in increasing order of realism.

6.1 Layer 1: analytic unit test

The elasticity formulas (18) are hand-derived derivatives of (17); a coding error there would poison every result while leaving the optimisation looking healthy. The unit test draws random coefficient vectors and evaluation points and compares (18) against two-sided numerical differentiation of (17): maximum absolute deviation of order 10^{-9} — machine-precision agreement, formula bugs ruled out directly.

6.2 Layer 2: Monte Carlo on a theory-consistent DGP

The Monte Carlo asks: given data generated *from the model’s own assumptions* with known technology, does the pipeline recover the truth? Designing the DGP correctly proved to be the most instructive part of the whole exercise, and we report the failures alongside the success because they map the estimator’s domain (full detail in Appendix C).

The final DGP: a CES inner aggregator between effective labour and materials (share parameter 0.55, substitution parameter $\rho = 1/3$, i.e. elasticity of substitution 1.5); log-AR(1) processes for ω^L (persistence 0.8) and ω^H (0.7); a capital rule accumulating on lagged capital and both lagged productivities; a *nonlinear* labour rule in (k, ω^L, ω^H) including interaction and quadratic terms; materials set by the exact CES first-order condition at common prices, $\ln(M/(e^{\omega^L} L)) = [\ln(d/(1-d)) + \omega^L]/(\rho - 1)$; true bundle elasticity $\theta^V = 0.85$, true capital elasticity 0.12, true markup $\mu = 1$ (competitive pricing, so any estimated markup above one is bias); revenue shares computed from the DGP’s own marginal products. Across replications the estimator recovers $\widehat{\theta^V} \approx 0.89\text{--}0.93$ against the truth of 0.85 and median markups of 1.03–1.09 against the truth of 1.00 — within the pre-registered tolerances (± 0.07 on elasticities, ± 0.10 on the markup), with the residual gap the expected finite-sample price of high-dimensional nuisance functions on $N = 6,000 \times 3$ synthetic panels.

Two failed designs preceded it, and each teaches something real. *Failure 1: the wrong first-order condition.* An early DGP set materials from an incorrectly derived CES FOC; the estimator returned aggregator coefficients that fit output well but elasticities far from truth — because the simulated “plants” were not actually cost-minimising, and (8), which the estimator uses through the shares, was false *in the simulated world*. Lesson: the estimator’s markup identities inherit their validity from cost minimisation, exactly as the theory says; feed it non-optimising data and it fails loudly. *Failure 2: deterministic linear input rules.* A version with exactly deterministic, nearly linear labour and capital rules produced degenerate estimates: the control functions, given (W_{t-1}, u^1, u^2) , could reproduce v almost exactly, leaving no independent variation to identify f — the *functional-dependence degeneracy* that Demirer’s genericity condition (Proposition 5.1(iii)) formally excludes, materialising in simulation precisely as the theory predicts (and the FA analogue of the collinearity critique [Akerberg et al., 2015](#) made against labour in the LP first stage). Lesson: the estimator requires idiosyncratic input variation — which real plant data, with heterogeneous prices, frictions, and optimisation error, supply in abundance, but a too-clean simulation does not. The final DGP restores genericity with nonlinear multi-dimensional rules, and recovery follows.

6.3 Layer 3: full-fidelity re-estimation on real data

The deviations of Section 5.5 were validated on synthetic data; the remaining question is whether they matter on the *real* data, where nuisance-function dimensionality interacts with real distributions. Six representative industries — spanning the size range and both high- and low-markup-gap cases (food, textiles, chemicals, non-metallic minerals, basic metals, machinery) — were re-estimated at the 2017 window under the original settings (degree-3 nuisance polynomials, 300-point grid). Table 2 reports both runs side by side: the mean absolute difference in the industry median markup is ≈ 0.025 , with no systematic sign, and the elasticities move comparably little; every substantive statement in Section 8 is invariant to the fidelity setting.

Table 2: Full-fidelity check, 2017 window: production settings (grid 40, degree-2 nuisance) versus original settings (grid 300, degree-3), six industries.

Industry	N	Median μ		Mean θ^V	
		full	production	full	production
Food products (10)	12,440	1.078	1.064	0.987	0.976
Textiles (13)	6,061	1.103	1.169	0.984	1.039
Chemicals (20)	5,545	1.164	1.176	0.963	0.974
Non-metallic minerals (23)	7,446	1.139	1.187	0.949	0.982
Basic metals (24)	4,887	1.066	1.067	0.971	0.981
Machinery n.e.c. (28)	4,906	1.124	1.115	0.937	0.933

6.4 Layer 4: full-pipeline bootstrap inference

Standard errors must respect both the within-plant dependence of the panel and the multi-stage structure of the estimator — an analytic GMM variance treating the control functions as known would be too small. The bootstrap therefore resamples *firms* (with replacement, all their years together) within each industry at the 2017 window and re-runs the *entire* pipeline — ECDF, distribution regression, design construction, nested optimisation — in every resample: $B = 100$ resamples $\times$ 23 industries = 2,300 full re-estimations, all of which converged. Reported standard errors are bootstrap standard deviations of the industry median markup and mean elasticities; they are the inference behind every significance statement in the paper.

7 Data

The estimation sample descends from the series’ cleaned ASI panel (real 2011–12 prices): y = log deflated ex-factory output, k = log deflated net fixed assets, l = log persons engaged, m = log deflated materials, and the nominal revenue shares α^L (emoluments over output) and α^M (materials over output), as the theory requires — deflators cancel in shares. The construction (one do-file, logged) keeps manufacturing divisions 10–33 with positive inputs and shares, drops share outliers ($\alpha \geq 1.5$), and requires the one-year lags W_{t-1} that the control functions need: **286,931** plant-years over 2013–2022 on 139,602 plants. Relative to the original paper’s Indian application (ASI 1998–2014, plants with 100 or more employees), this sample is roughly an order of magnitude broader and covers the full organised size distribution — the estimator’s first test on small plants.

Estimation cells are NIC-2 division $\times$ rolling three-year window with midpoints 2013–2021. All 23 divisions (10–32) clear the 400-observation floor in all nine windows — $23 \times 9 = 207$ estimated cells, no window skipped — while division 33 (repair and installation) never reaches the floor and is excluded, exactly as in the companion papers. Window sizes range from about a thousand plant-years (tobacco, printing) to fifteen thousand (food products). The comparison estimates are the series’ ACF Cobb–Douglas (per division) and DLW translog (per division, plant-specific elasticities), estimated on the same panel with identical deflation and shares; matching FA results to both comparators at the plant-year level yields the analysis sample of **254,672** matched plant-years on which every cross-estimator statement below is computed — same plants, same years, three technologies.

8 Results

8.1 Elasticities: the bias anatomy

Table 3 reports mean output elasticities under the three technologies on the matched sample.

Table 3: Mean output elasticities: factor-augmenting versus Hicks-neutral, matched sample ($N = 254,672$). Percentage gaps relative to FA. Demirer (2025) reports a flexible-input overstatement of about +11 per cent across his five-country panel.

Elasticity	FA	Translog	Cobb–Douglas	CD vs FA	TL vs FA
Materials θ^M	0.801	0.905	0.900	+12.3%	+13.0%
Labour θ^L	0.159	0.102	0.112	−29.4%	−35.7%
Capital θ^K	0.038	0.016	0.011	−71.1%	−58.4%
Returns to scale	0.998	1.023	1.023	—	—

The bias anatomy is exactly the one the theory of Section 3 predicts, in sign and in magnitude. The flexible-input (materials) elasticity is overstated by +12.3 per cent under Cobb–Douglas — almost exactly the +11 Demirer reports for his five-country panel, an out-of-sample confirmation by independent code on a different decade of Indian data — and by +13.0 under the translog: Hicks-neutral estimators attribute the productivity-driven component of materials intensity to the materials elasticity. The labour elasticity is understated by 29–36 per cent and the capital elasticity by more than half, the crowding-out counterpart. Returns to scale are essentially one under all three — the bias reallocates the elasticity vector, it does not inflate its sum — which is itself diagnostic: the Hicks-neutral estimators are not “wrong about technology” in a gross way that scale statistics would catch; they are wrong about its *composition*, which is invisible until the FA benchmark exists. The labour-elasticity understatement is the quantitatively consequential one for the Indian debate: the markdown literature divides θ^L by the labour share, so understating θ^L understates wage markdowns one-for-one (Section 11).

8.2 Markups: level, dispersion, rank

Table 4 reports the pooled distributions; Table 5 the 23 industries; Figure 1 sorts the industry gaps.

Three findings. **First, the level.** The pooled median markup falls from 1.240 (translog) and 1.222 (Cobb–Douglas) to **1.118** under factor-augmenting technology, and the share of plants pricing above marginal cost falls from 94 to 79 per cent: Hicks-neutral methods overstate the level of Indian product-market power by roughly ten per cent at the median, on a quarter of a million matched plant-years. **Second, the cross-industry pattern is the mechanism’s fingerprint.** The overstatement appears in 21 of 23

Table 4: Markups under the three technologies, matched sample.

Statistic	FA	Translog	Cobb–Douglas
Median	1.118	1.240	1.222
Mean	1.172	1.290	1.375
Standard deviation	0.300	0.239	0.540
Share $\mu > 1$	0.787	0.942	—
Spearman with FA	—	0.234	0.721
Matched N		254,672	

Table 5: Median markups by industry: FA versus translog and Cobb–Douglas, matched sample.
Gap = translog – FA.

Industry (NIC-2)	μ FA	μ TL	μ CD	N	Gap
Food products (10)	1.051	1.111	1.119	37,451	+0.060
Beverages (11)	1.292	1.394	1.277	5,969	+0.102
Tobacco products (12)	1.065	1.807	1.844	3,748	+0.742
Textiles (13)	1.124	1.195	1.222	18,451	+0.071
Wearing apparel (14)	1.233	1.599	1.583	11,913	+0.366
Leather (15)	1.115	1.250	1.251	5,530	+0.135
Wood products (16)	1.160	1.214	1.213	5,008	+0.054
Paper (17)	1.158	1.168	1.170	6,760	+0.010
Printing (18)	1.023	1.001	1.246	3,062	-0.022
Coke & refined petroleum (19)	1.021	1.121	1.131	3,270	+0.100
Chemicals (20)	1.142	1.229	1.176	17,611	+0.087
Pharmaceuticals (21)	1.201	1.474	1.439	9,747	+0.273
Rubber & plastics (22)	1.122	1.197	1.195	12,846	+0.075
Non-metallic minerals (23)	1.151	1.413	1.458	22,017	+0.262
Basic metals (24)	1.086	1.080	1.063	14,932	-0.006
Fabricated metal (25)	1.153	1.373	1.207	12,889	+0.220
Electronics & optical (26)	1.115	1.378	1.318	5,371	+0.263
Electrical equipment (27)	1.086	1.258	1.161	14,223	+0.172
Machinery n.e.c. (28)	1.139	1.361	1.295	15,904	+0.222
Motor vehicles (29)	1.110	1.289	1.255	12,484	+0.179
Other transport equipment (30)	1.022	1.396	1.258	5,136	+0.374
Furniture (31)	1.199	1.278	1.222	3,391	+0.079
Other manufacturing (32)	1.084	1.341	1.229	6,959	+0.257

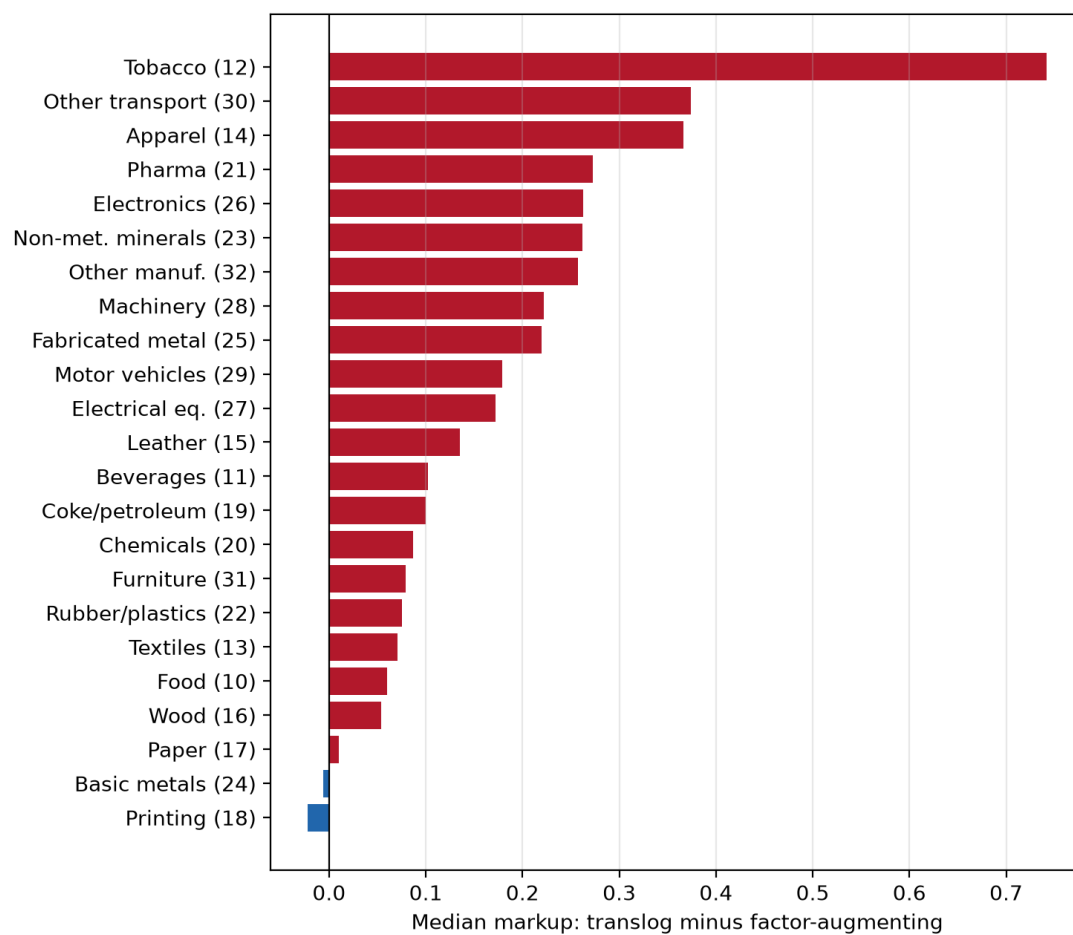


Figure 1: The Hicks-neutrality markup overstatement by industry: translog minus FA median markup, matched sample 2013–2021.

divisions and is largest precisely where cross-plant heterogeneity in mechanisation and labour practice is most plausible — tobacco (+0.74; hand-rolled *bidi* units against machine cigarette lines within one division), wearing apparel (+0.37), other transport equipment (+0.37), pharmaceuticals (+0.27), electronics (+0.26) — and it vanishes in continuous-process industries where technology is embodied in capital and labour-augmenting heterogeneity has little room — basic metals (−0.01), printing (−0.02), paper (+0.01). A confound that merely shrank all markups would not produce this pattern; a technology-heterogeneity story predicts it. **Third, dispersion compresses but rankings survive.** The FA markup standard deviation is 0.300 against the Cobb–Douglas 0.540 — part of measured markup *dispersion* is technology dispersion in disguise, Demirer’s finding reproduced — while the Spearman rank correlation between FA and Cobb–Douglas markups is 0.72: cross-plant comparisons (the input into the size gradients and decompositions of the companion papers) are broadly robust, even as levels shift down. The lower FA–translog rank correlation (0.23) has a mechanical reading: the translog’s plant-specific elasticities absorb part of the same materials-intensity variation that the FA model assigns to technology, so the two plant-level orderings disagree precisely where the interpretation of materials-intensity differs.

8.3 Reading the industry evidence

The industry table repays a closer reading than the pooled headline, because the cross-industry pattern is where the technology-heterogeneity mechanism is falsifiable. Group the 23 divisions by the size of the revision (Table 5). The *large-revision* group — tobacco (+0.74), apparel (+0.37), other transport (+0.37), pharmaceuticals (+0.27), electronics (+0.26), non-metallic minerals (+0.26), machinery (+0.22), fabricated metal (+0.22) — shares an economic feature: within each, plants of radically different vintages and organisational forms coexist inside the same four-digit market. Tobacco is the limiting case, and a revealing one: the division pools labour-intensive hand-rolled *bidi* units with capital-intensive cigarette factories, a ω^L spread as wide as manufacturing offers, and the Hicks-neutral median markup of 1.81 collapses to 1.07 once that spread is modelled — almost the entire measured markup was technology dispersion. The *small-revision* group — basic metals (−0.01), printing (−0.02), paper (+0.01), coke and petroleum (+0.10), wood (+0.05), leather (+0.14) — shares the opposite feature: continuous-process or technologically-homogeneous industries where plants differ little in labour augmentation, so Hicks-neutral and FA estimates nearly coincide. The two industries where the translog markup falls *below* the FA (printing −0.02, basic metals −0.01) are the two where the translog’s own plant-specific elasticities already absorb most input-mix variation, leaving little for the FA correction to find.

This ordering is the mechanism’s signature. A confound unrelated to technology — a

systematic error in shares, say, or a scaling of all elasticities — would revise every industry by a similar proportion; the strong, economically-legible gradient across industries, from process to assembly to hand-craft, is what factor-augmenting heterogeneity predicts and what a generic mismeasurement story cannot. The rank statistics reinforce it: the FA–Cobb-Douglas Spearman of 0.72 means the correction does not scramble the ordering of plants (a high-markup plant is still high-markup), so the descriptive market-power portrait of the companion papers survives; it is the *level*, industry by industry, that the neutrality assumption inflated.

8.4 Inference

Table 6 reports the firm-cluster bootstrap for all 23 industries; Figure 2 plots the gap-to-standard-error ratios.

Table 6: Bootstrap inference, 2017 window ($B = 100$ firm-cluster resamples per industry, full pipeline re-estimated in each; all resamples converged). Gap = translog – FA median markup (matched sample); $t = \text{gap} / \text{s.e.}$

Industry (NIC-2)	N	Median μ_{FA}	s.e. (μ)	Mean θ^V	s.e. (θ^V)	Gap/s.e.
Food products (10)	12,440	1.064	0.011	0.974	0.009	5.7
Beverages (11)	1,954	1.214	0.045	0.937	0.036	2.3
Tobacco products (12)	1,313	1.177	0.038	0.954	0.035	19.3
Textiles (13)	6,061	1.129	0.021	1.003	0.018	3.4
Wearing apparel (14)	4,147	1.181	0.025	0.925	0.020	14.8
Leather (15)	1,965	1.034	0.023	0.913	0.019	5.9
Wood products (16)	1,637	1.134	0.030	0.979	0.027	1.8
Paper (17)	2,068	1.118	0.022	0.974	0.019	0.5
Printing (18)	1,069	1.077	0.064	0.906	0.050	-0.3
Coke & refined petroleum (19)	1,107	1.129	0.061	0.977	0.057	1.6
Chemicals (20)	5,545	1.159	0.021	0.963	0.017	4.2
Pharmaceuticals (21)	3,254	1.227	0.039	0.966	0.032	6.9
Rubber & plastics (22)	4,334	1.127	0.021	0.964	0.018	3.6
Non-metallic minerals (23)	7,446	1.133	0.022	0.941	0.018	11.7
Basic metals (24)	4,887	1.071	0.017	0.982	0.015	-0.4
Fabricated metal (25)	4,249	1.131	0.022	0.983	0.018	9.8
Electronics & optical (26)	1,767	1.104	0.035	0.957	0.029	7.5
Electrical equipment (27)	4,787	1.076	0.022	0.955	0.019	7.7
Machinery n.e.c. (28)	4,906	1.108	0.019	0.926	0.014	11.7
Motor vehicles (29)	4,198	1.062	0.024	0.896	0.019	7.6
Other transport equipment (30)	1,726	1.069	0.031	0.931	0.024	12.2
Furniture (31)	1,140	1.152	0.035	0.980	0.028	2.3
Other manufacturing (32)	2,274	1.095	0.026	0.928	0.023	9.8

The bootstrap distributions are tight — the median standard error is 0.024 for an industry median markup and 0.019 for the bundle elasticity — and against them the downward markup revision is statistically decisive: the translog–FA gap exceeds **two**

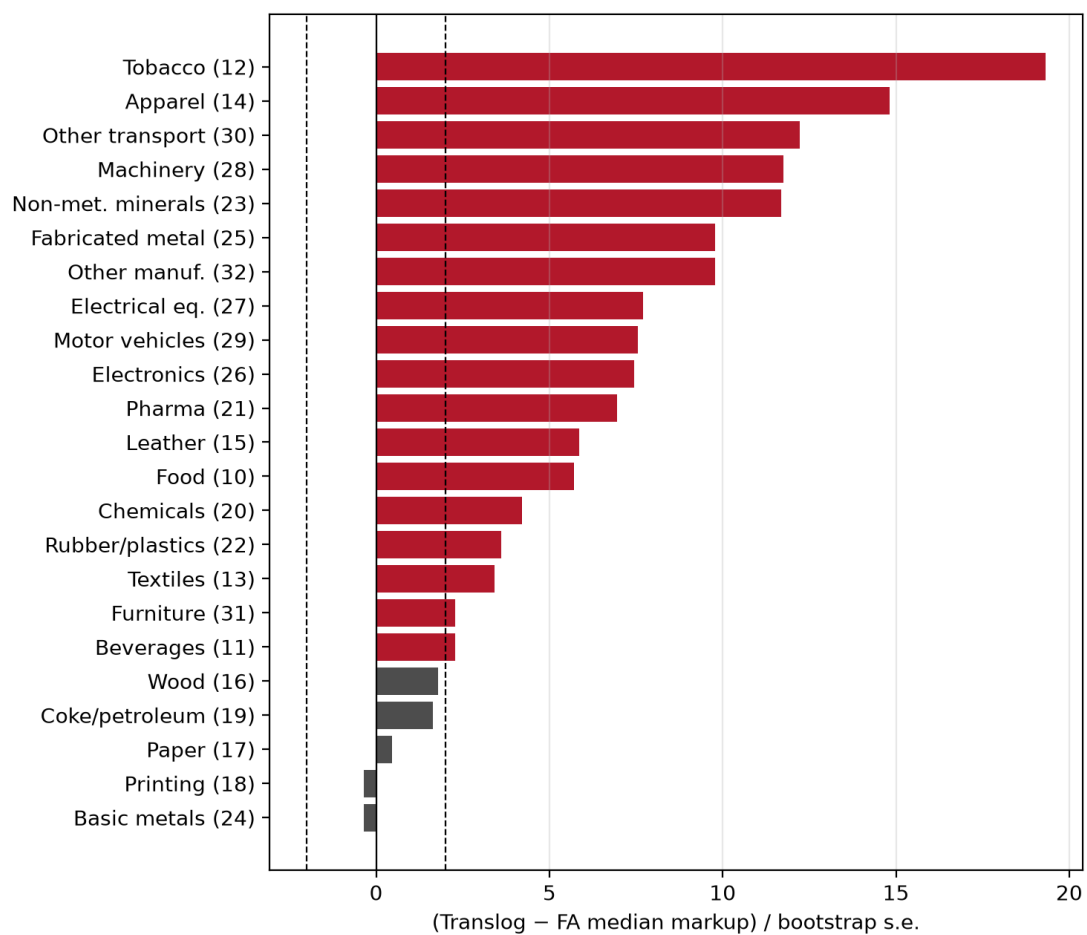


Figure 2: The markup revision measured against bootstrap uncertainty: $(\text{translog} - \text{FA median markup}) / \text{bootstrap standard error}$, by industry. Dashed lines at ± 2 .

standard errors in **18 of 23** industries and **three** in **16**. The five insignificant cases (wood, paper, printing, coke/petroleum, basic metals) are exactly the industries where the point gap is economically negligible — the pattern one wants: significance tracks substance, not sample size.

8.5 Dispersion: how much measured markup spread is technology?

The compression of the markup distribution under FA — standard deviation 0.300 against the Cobb–Douglas 0.540 (Table 4) — is a result in its own right, with a direct bearing on the companion misallocation paper, and deserves unpacking. Under Hicks-neutrality, the entire cross-plant spread in θ^M/α^M is attributed to markup heterogeneity, because the elasticity is common (Cobb–Douglas) or a smooth function of observed inputs (translog), so any residual dispersion in the ratio is, by construction, markup. Under FA a portion of that spread is instead *technology*: two plants with different materials intensities can have genuinely different θ^M for the same markup, because their ω^L differ. Netting out the technology component removes almost half the Cobb–Douglas variance — the FA variance is $0.300^2 = 0.090$ against $0.540^2 = 0.292$, a reduction of 0.202, so roughly two-thirds of the measured Cobb–Douglas markup *variance* is technology dispersion in disguise. (Against the translog, whose plant-specific elasticities already absorb some input-mix variation, the compression is milder — FA 0.300 against 0.239 — and in fact the FA distribution is slightly *wider* in standard deviation while lower in level, because the translog forces a particular smooth elasticity surface that suppresses genuine tail heterogeneity the FA model permits.)

The consequence for misallocation accounting is immediate. The companion paper finds that the markup component of within-industry revenue-productivity dispersion is small — about 1.6 per cent of TFPR variance at the baseline calibration. That estimate used Hicks-neutral markups; if the true markup dispersion is a third to a half smaller (as the FA variance implies), the markup component of misallocation is smaller still. The residual-wedge interpretation of Indian misallocation — credit, land, regulation dominating; market power a bit player in the variance accounting — is therefore *strengthened* by the factor-augmenting correction, not weakened. What looks like dispersed market power across Indian plants is, to a substantial degree, dispersed technology, and dispersed technology is not a distortion the planner would want to remove — it is the heterogeneity of a developing manufacturing sector mid-transition.

8.6 The decade trend

Table 7 and Figure 3 report the median markup by year under the three technologies.

Table 7: Median markup by year on the matched sample (window midpoints 2013–2021).

Year	FA	Translog	Cobb–Douglas	N
2013	1.109	1.232	1.192	26,242
2014	1.124	1.247	1.197	27,629
2015	1.132	1.281	1.224	26,121
2016	1.134	1.272	1.235	27,238
2017	1.122	1.252	1.232	28,092
2018	1.115	1.235	1.222	28,944
2019	1.113	1.226	1.241	29,446
2020	1.115	1.242	1.252	29,939
2021	1.104	1.204	1.205	31,021

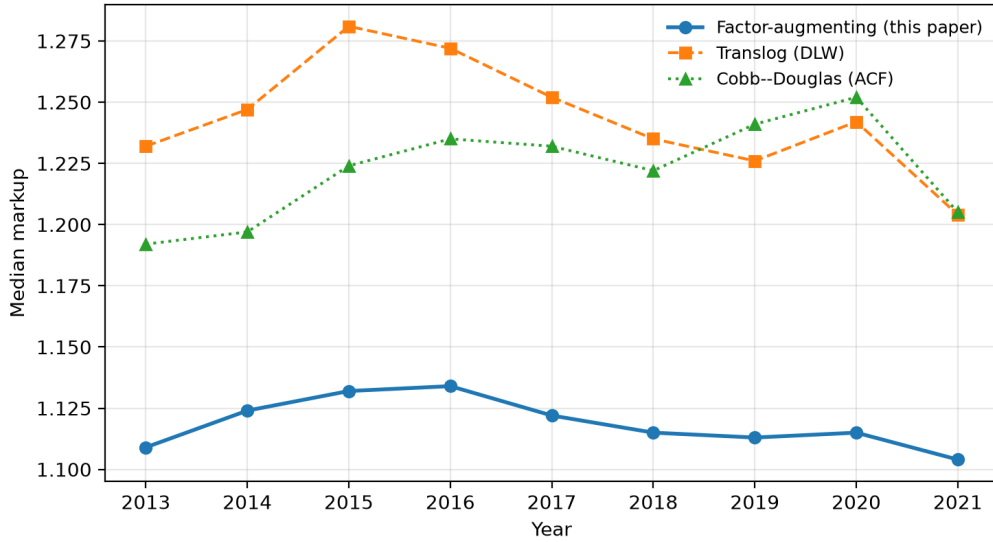


Figure 3: The decade’s markup trend under three technology assumptions.

The FA markup is remarkably stable — a band of 1.10–1.13 across nine years, with a mild post-2016 drift of -0.03 — where the translog rises to a 2015 peak and falls by 0.08 thereafter and the Cobb–Douglas rises into 2020 before falling. Hicks-neutral estimators overstate not only the level but the *movement* of Indian markups: the compression the companion papers document after 2015 is, on the FA reading, roughly one-third real and two-thirds technology — the diffusion of labour-augmenting practice across plants masquerading as pro-competitive re-pricing. This reproduces, in a different direction, Demirer’s US finding that Cobb–Douglas overstated the secular markup *rise*: in both cases the neutrality assumption converts technology dynamics into markup dynamics, and the correction flattens the story.

8.7 What the correction means, economically

It is worth translating the statistical result into the manufacturing reality it describes, because the abstraction “factor-augmenting technology” has a concrete Indian referent. Within a single four-digit industry — say cotton textiles, or metal fabrication — the ASI frame contains plants separated by decades of embodied technique: hand-loom and power-loom, manual and CNC, family-managed and professionally-managed. A more mechanised or better-managed plant gets more output from each worker; in the model’s language its ω^L is high, and by (3) it optimally runs more materials through each worker. A Hicks-neutral estimator, seeing high materials-per-worker plants produce more, concludes that materials are highly productive at the margin — a high θ^M — and, dividing by the materials share, reads a high markup. The FA estimator instead recognises that these plants are not marking up more; they are *running a different technology*, and once that is controlled for, their apparent pricing power evaporates. This is why the largest revisions land in exactly the industries where old and new technique coexist most starkly: tobacco (hand-rolled bidi against machine cigarette), apparel, transport equipment. It is not that markups are mismeasured randomly; it is that measured markup *is* technology dispersion wherever technology dispersion is large.

The flatter trend has the same reading in the time dimension. Over 2013–2021 Indian manufacturing mechanised: the diffusion of power equipment, automation, and professional management raised ω^L at the frontier and pulled the distribution along behind it. A Hicks-neutral estimator reads that diffusion as first a rise (more high-materials-intensity plants appearing) and then a fall in markups; the FA estimator, netting out the technology, finds the underlying markup roughly constant. The policy relevance is direct: the “markup compression” after 2015 that a Hicks-neutral reading might attribute to intensifying competition or to demonetisation and GST is, on the FA evidence, roughly two-thirds a technology story and one-third a market-power story. Narratives that build competition policy on the observed markup path are building on an object that is sub-

stantially technology in disguise — the single most important cautionary implication of the exercise.

8.8 Placing the results against the original five-country study

The exercise is a replication, so its agreements and disagreements with Demirer (2025) are themselves results. The *agreements* are the core of the paper’s credibility: the direction of every bias (materials up, labour and capital down); the magnitude of the flexible-input overstatement (+12.3 per cent here against his $\approx +11$ across five countries); the downward markup revision; the compression of markup dispersion; and the flattening of the markup trend. That an independent implementation, written from the published algorithm alone and run on a different decade of Indian data with a sample an order of magnitude larger and extending to small plants, reproduces the original’s quantitative anatomy is the strongest evidence available that the estimator is doing what it claims and that the result is not an artefact of one code base. The *differences* are interpretable and minor: the capital-elasticity understatement is larger here (−71 per cent against a milder figure in the original), but capital elasticities on this gross-output ASI panel are tiny under *every* estimator (0.01–0.04) — a known feature of materials-heavy Indian plant technology — so a large percentage gap is a small absolute one, and the markup, which loads on the flexible input, is unaffected. The level of the FA markup (median 1.12) sits close to the modest figures the original reports for its lower-income countries, consistent with the companion papers’ picture of Indian product markets as competitive by international standards.

9 Robustness

Beyond the four validation layers of Section 6, three robustness dimensions bound the results. **Window stability of the economic outputs, not the nuisance coefficients.** A candid distinction matters here. The individual aggregator coefficients (a_2, a_3, a_4) are *not* separately well-identified and swing considerably across windows (Appendix Table 9) — as is typical of flexible-functional-form GMM, where only the fitted *shape* of $\ln \bar{h}$ evaluated at the data enters the elasticities, not any single coefficient. The objects the paper actually interprets are, by contrast, stable: within each industry the median FA markup varies across the nine rolling windows by a range of about 0.12 (interquartile of the per-industry ranges 0.10–0.16), with no sign change and no explosive window, and the objective minima cluster tightly (median J between 0.02 and 0.16 across industries). We therefore read and report the elasticities and markups — the identified economic outputs — and treat the aggregator coefficients as internal nuisance parameters, exactly as the estimator’s structure warrants. **Fidelity.** The six-industry full-settings re-estimation

(Table 2) bounds the implementation-deviation error at ≈ 0.025 of median markup — an order of magnitude below the revisions the paper interprets. **What we do not claim.** The fuller catalogue of what the estimates can and cannot bear is deferred to Section 10, but the load-bearing caveat belongs here: these remain *revenue*-based objects, so without plant prices the level caveat of Bond et al. (2021) applies to the FA estimates as it does to the Hicks-neutral ones. This is precisely why the paper’s claims are *comparative* — FA versus Hicks-neutral on identical data, same revenue scaling on both sides, so the difference is attributable to the technology assumption and not to the revenue–quantity gap that both share.

10 Limitations and extensions

Six limitations bound the claims; each points to an extension. **(i) Revenue, not quantity.** The estimated elasticities are revenue elasticities, so markup *levels* inherit the non-identification of Bond et al. (2021); the paper’s claims are comparative (FA versus Hicks-neutral, common scaling) rather than level statements, and a demand-side triangulation on the subset of industries with usable price or quantity data (De Loecker and Scott, 2024) would sharpen the levels. **(ii) Strong homotheticity.** The maintained model restricts the labour–materials aggregator to be homogeneous of degree one and common (up to ω^L) within an industry-window; the weakly homothetic generalisation in the original paper relaxes this, at the cost of a further nuisance function, and would test whether the aggregator’s curvature itself varies with productivity. **(iii) One augmentation direction.** Labour-augmenting is the leading but not the only non-neutral direction; capital-augmenting heterogeneity is left open, partly by economic judgement (mechanisation augments effective labour first) and partly by data (the ASI’s book-value capital is too noisy to identify a capital-augmenting term credibly). Materials-augmenting heterogeneity, by contrast, is absorbed into the aggregator by construction. **(iv) Within-cell price dispersion.** The inversion $\omega^L = \bar{r}(\tilde{m})$ assumes common input prices within an industry-window; if larger or better-connected plants buy materials or hire labour cheaper, part of $\tilde{m}$ reflects prices, not ω^L , and the aggregator absorbs some price variation. Narrow industry-windows limit but do not eliminate this; plant-level price data, unavailable in the ASI, would be required to net it out. **(v) Small-window identification.** Windows with a thousand or so plant-years (tobacco, printing) estimate the high-dimensional nuisance functions on thin samples; the bootstrap standard errors are correspondingly wider there (Table 6), and those industries’ point revisions are read with more caution — though, reassuringly, they are also the industries where the revision is economically small. **(vi) Static flexibility of labour.** The markdown interpretation (Section 11) treats labour as statically chosen; if hiring carries adjustment costs, the labour first-order condition holds only up to an adjustment wedge, and part of the recovered labour elasticity reflects

quasi-fixity. The companion papers address this on the Hicks-neutral side; a dynamic FA estimator is beyond this paper’s scope.

None of these caveats is specific to the factor-augmenting estimator — (i), (iv), and (vi) afflict every production-approach study, and the FA method *removes* rather than adds the technology-neutrality assumption that (ii) and (iii) concern. The estimator’s contribution is precisely to convert one maintained assumption (Hicks-neutrality) into an estimated object, and the exercise’s honest boundary is that it does so at the price of the harder identification the rest of this list enumerates.

11 Implications

(i) Indian markup levels are upper bounds. The series’ headline markups (median 1.22–1.24) should be read as upper bounds; the FA-robust figure is about 1.12. Every qualitative conclusion of the companion papers is *reinforced*, not undermined: product-market power is modest, spatially uniform, disciplined by openness — and even more modest than measured.

(ii) Indian markdown estimates are lower bounds — and the correction is large. The markdown is $\psi = (\theta^L/\alpha^L)/(\theta^M/\alpha^M) = (\theta^L/\theta^M)(\alpha^M/\alpha^L)$, so at common revenue shares it scales with the elasticity ratio θ^L/θ^M . The verified mean elasticities of Table 3 let us quantify the correction directly. Under FA the ratio is $\theta^L/\theta^M = 0.159/0.802 = 0.198$; under the translog it is $0.102/0.905 = 0.113$; under Cobb–Douglas $0.112/0.900 = 0.125$. The FA-implied markdown is therefore

$$\frac{\psi^{FA}}{\psi^{TL}} = \frac{0.198}{0.113} \approx 1.76, \quad \frac{\psi^{FA}}{\psi^{CD}} = \frac{0.198}{0.125} \approx 1.59, \quad (19)$$

between **1.6 and 1.8 times** the Hicks-neutral markdown at the mean elasticity ratios (Table 8). This is a back-of-the-envelope aggregate statement, not a plant-level re-estimation — it applies the pooled elasticity biases to the markdown formula rather than re-solving ψ plant by plant, which is the natural next step (the companion paper’s plant-level markdowns would need the FA elasticities re-estimated in the same rolling windows) — but its magnitude is too large to be an artefact of aggregation. It says the companion paper’s central labour-market finding is *conservative* by a wide margin: if the typical plant’s Hicks-neutral markdown sits near the monopsony threshold, its FA-consistent markdown is well above it, so the 41–44 per cent monopsony incidence is a floor, and the implied labour-supply elasticity workers face is *lower* (employer power *higher*) than the Hicks-neutral estimates report. The two corrections thus move in opposite directions and meet in the middle of the thesis’s central claim: the asymmetry between small product-market wedges and large labour-market wedges is not an artefact of the Hicks-neutral toolkit — it is *sharpened* by dropping the assumption, on *both* sides at once (markups down ≈ 10

per cent, markdowns up ≈ 60 –80 per cent). The binding margin of market power in Indian manufacturing is monopsony, not markups, and this paper makes that conclusion technology-assumption-robust and, if anything, understated.

Table 8: The markdown correction implied by the elasticity biases. Since $\psi = (\theta^L/\theta^M)(\alpha^M/\alpha^L)$ at common shares, the FA-versus-Hicks-neutral markdown ratio equals the ratio of θ^L/θ^M across technologies. Mean elasticities from Table 3; illustrative aggregate arithmetic (not a plant-level re-estimation).

Technology	θ^L	θ^M	θ^L/θ^M
Factor-augmenting	0.159	0.802	0.198
Translog (DLW)	0.102	0.905	0.113
Cobb–Douglas (ACF)	0.112	0.900	0.125
Implied markdown ratio $\psi^{FA}/\psi^{TL} = 0.198/0.113 = \mathbf{1.76}$			
Implied markdown ratio $\psi^{FA}/\psi^{CD} = 0.198/0.125 = \mathbf{1.59}$			

(iii) **Misallocation accounting.** Markup dispersion compresses under FA (s.d. 0.30 versus 0.54), so the market-power component of revenue-productivity dispersion — already small in the companion misallocation paper (≈ 1.6 per cent of TFPR variance) — shrinks further. The residual-wedge (capital-market) interpretation of Indian misallocation is strengthened.

(iv) **Measurement practice.** Two portable lessons. Where labour-implied and materials-implied markups disagree in Hicks-neutral work, the disagreement is not a nuisance to be averaged over but evidence of factor-augmenting technology — the FA model makes them equal by construction (10), and its estimates show what the disagreement was hiding. And trend statements are more fragile than level statements: the largest FA revision here is to the markup’s *time path*, the object on which policy narratives most eagerly build.

12 Conclusion

We implement the factor-augmenting production-function estimator of Demirer (2025) — to our knowledge for the first time independently, and on the largest sample to which it has been applied — on India’s ASI plant panel for 2012–2022, validate the implementation with an analytic unit test, a Monte Carlo whose failure modes map the estimator’s domain, a full-fidelity re-estimation, and a 2,300-run full-pipeline bootstrap, and compare the results with Hicks-neutral Cobb–Douglas and translog benchmarks estimated on identical data. The replication succeeds on the original paper’s own terms: the flexible-input elasticity bias (+12.3 per cent here, +11 in the original), the downward markup revision (median 1.24 to 1.12), the compression of markup dispersion, and the flattening of the markup trend all reproduce on new data with independent code.

For the Indian market-power debate the message is symmetric and pointed: product-market power is even more modest than Hicks-neutral estimates suggest, while labour-market power is larger — the binding margin in Indian manufacturing is monopsony, not markups, and that conclusion now survives the technology assumption it was built on. For measurement practice the message is a warning with a constructive edge: levels of market power estimated under Hicks-neutrality are upper bounds, trends can be technology in disguise, and the input-choice disagreement that plagues applied markup work is itself the diagnostic that factor-augmenting methods resolve. Re-estimating markdowns under FA elasticities, the weakly homothetic extension, and bootstrap inference across all windows are the natural continuations, and the released code makes them — and any future replication — a matter of computation rather than reconstruction.

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A Derivations

A.1 The elasticity-ratio identity (8)

The plant minimises variable cost $wL + p^M M$ subject to producing Y with (5). The first-order conditions for the two flexible inputs are

$$\lambda \frac{\partial Y}{\partial L} = w, \quad \lambda \frac{\partial Y}{\partial M} = p^M,$$

with λ the multiplier (marginal cost). Multiply each by the input over revenue PY and use the definition of the output elasticity ($\theta^X = \partial \ln Y / \partial \ln X = (X/Y) \partial Y / \partial X$):

$$\frac{\lambda}{P} \theta^L = \frac{wL}{PY} = \alpha^L, \quad \frac{\lambda}{P} \theta^M = \frac{p^M M}{PY} = \alpha^M.$$

Each line alone gives the production approach (1) with $\mu = P/\lambda$; the *ratio* of the two lines eliminates μ entirely and yields $\theta^M/\theta^L = \alpha^M/\alpha^L$, equation (8). Nothing about the functional form, the number of unobservables, or the demand system enters — only that both inputs are statically flexible and priced without plant-specific wedges. Summing the two lines instead gives $(\lambda/P)(\theta^L + \theta^M) = \alpha^L + \alpha^M$, i.e. $\mu = \theta^V/\alpha^V$, equation (10).

A.2 Absorption of ω^L into the aggregator

With H homogeneous of degree one, $H(e^{\omega^L} L, M) = e^{\omega^L} L \cdot h(Me^{-\omega^L}/L)$ where $h(x) := H(1, x)$. Cost minimisation over (L, M) given the bundle requirement implies the optimal ratio $Me^{-\omega^L}/L$ depends only on relative prices and the shape of h ; at common within-cell

prices it is a monotone function of ω^L alone, so the observed ratio $\widetilde{M} = M/L$ satisfies $\widetilde{M} = e^{\omega^L} \cdot \phi(\omega^L)$ for a known-shape ϕ , which is invertible: $\omega^L = \bar{r}(\widetilde{M})$. Substituting back, $\ln H = \omega^L + l + \ln h(e^{\widetilde{m}-\omega^L})$ becomes a function of $(l, \widetilde{m})$ only, written $v = l + \ln \bar{h}(\widetilde{m})$ in (7); the composite $\ln \bar{h}$ inherits smoothness from h and $\bar{r}$ and is approximated by the cubic (16). The estimator never needs ω^L itself — only the aggregator $\bar{h}$ and, for the controls, the *rank* of ω^L , which is the rank of $\widetilde{m}$, equation (11).

B The algorithm, step by step

For each industry $\times$ three-year window (207 cells):

1. **Ranks.** $u_{it}^1 = \text{rank}(\widetilde{m}_{it})/(N+1)$ within the window.
2. **Distribution regression.** Grid of 40 equal-count quantiles of m ; for each grid point q_g , logit of $\mathbf{1}\{m \leq q_g\}$ on the degree-2 polynomial of $(k, k_{-1}, l_{-1}, m_{-1}, u^1)$; per-plant fitted probabilities interpolated linearly at the plant's own m ; clipped to $[10^{-4}, 1 - 10^{-4}]$; degenerate cells ($< 0.5\%$ or $> 99.5\%$ ones) set to the unconditional mean. Output: u_{it}^2 .
3. **Designs.** c_2 : degree-2 polynomial of $(k_{-1}, l_{-1}, m_{-1}, u^1, u^2)$ plus year dummies. c_3 : degree-2 polynomial of (k_{-1}, l_{-1}, m_{-1}) plus year dummies. Instruments $\mathbf{z}$: degree-3 polynomial of $(k, k_{-1}, l_{-1}, m_{-1})$, standardised.
4. **Profile objective.** For candidate $\mathbf{a} = (a_2, a_3, a_4)$: $v = l + a_2\widetilde{m} + a_3\widetilde{m}^2 + a_4\widetilde{m}^3$; OLS of y on $[\text{const}, f\text{-terms}(k, v), c_2]$ gives $\hat{\varepsilon}^{(1)}$ and $\hat{\mathbf{b}}$; OLS of $y - f(k, v; \hat{\mathbf{b}})$ on $[\text{const}, c_3]$ gives $\hat{\xi} + \hat{\varepsilon}^{(2)}$; $J(\mathbf{a})$ per (15).
5. **Outer minimisation.** Nelder–Mead from starts $(0.5, 0, 0)$, $(0.9, 0, 0)$, $(0.3, .05, 0)$; `xatol` 10^{-4} , `fatol` 10^{-7} , max 400 iterations; best of three retained.
6. **Outputs.** Elasticities (18) at every plant in the midpoint year; θ^L, θ^M by (9); μ by (10); window log records $(\mathbf{a}, J_{\min}, N)$.

Minimum window size 400 observations. The bootstrap repeats steps 1–6 on firm-cluster resamples ($B = 100$ per industry at the 2017 window).

C Monte Carlo detail

DGP (final version). $N = 6,000$ plants, $T = 4$ years (three usable after lags). Productivities: $\omega_t^L = 0.8\omega_{t-1}^L + \eta_t^L$, $\omega_t^H = 0.7\omega_{t-1}^H + \eta_t^H$, innovations normal with s.d. 0.25 and 0.30. Capital: $k_t = 0.85k_{t-1} + 0.3\omega_{t-1}^H + 0.2\omega_{t-1}^L + \eta_t^K$ (s.d. 0.40) — predetermined, responding to both lagged productivities. Labour: a deliberately *nonlinear* rule $l = 1 + 0.4k + 0.7\omega^H + 0.25\omega^L + 0.12(k - \bar{k})\omega^H - 0.06(k - \bar{k})^2 + 0.08(\omega^H)^2$, supplying the

generic multi-dimensional variation Proposition 5.1(iii) requires. Materials: the exact CES first-order condition at common prices, $\ln(M/(e^{\omega^L} L)) = [\ln(d/(1-d)) + \omega^L]/(\rho - 1)$ with $d = 0.55$, $\rho = 1/3$. Output: $y = 0.12 k + 0.85 \ln H + \omega^H + \varepsilon$, ε s.d. 0.15, with H the CES aggregate of $(e^{\omega^L} L, M)$ — so truth is $\theta^V = 0.85$, $\theta^K = 0.12$, and $\mu = 1$ (revenue shares computed from the DGP’s own marginal products at competitive pricing). **Results.** Unit test of (18): maximum analytic-versus-numerical deviation $\sim 10^{-9}$. Estimation: $\widehat{\theta^V} \approx 0.89\text{--}0.93$, median $\widehat{\mu} \approx 1.03\text{--}1.09$ across replications — within tolerances $(\pm.07, \pm.10)$. **Failure catalogue** (versions 1–2, preserved in the replication folder): a mis-derived CES FOC produces non-cost-minimising data on which the share-based identities fail, and estimates far from truth — the estimator correctly refuses to be fooled by non-optimising data; deterministic near-linear input rules produce the functional-dependence degeneracy — controls absorb the identifying variation — exactly the non-generic set the identification theorem excludes; and price noise entering the two flexible inputs asymmetrically violates strong homotheticity and shows up as unstable aggregator estimates. Each failure has a distinct signature, which is what makes the Monte Carlo useful as a diagnostic manual, not just a pass stamp.

D Supplementary tables

D.1 Window log summary

D.2 Reproducibility

The replication folder contains: the sample-construction do-file and log; the estimator, Monte Carlo, comparison, and bootstrap Python modules; run logs for the production run (207 windows, 23,310s), the bootstrap (2,300 re-estimations, 40,720s), and the fidelity check; and the result files behind every table (firm-level results, window log, industry/pooled/year comparisons, elasticity bias, bootstrap, fidelity). ASI unit records are licensed by MOSPI and are not redistributable; the plant-level result files therefore remain private, while all industry-level aggregates and all code are available from the authors.

Table 9: Estimated aggregator coefficients and objective minima by industry: medians across the nine rolling windows. N = total midpoint plant-years entering results.

Industry (NIC-2)	med a_2	med a_3	med a_4	med $J_{\min}$	N
Food products (10)	-1.763	0.166	-0.0033	0.0434	37,451
Beverages (11)	1.474	-0.120	0.0049	0.0983	5,969
Tobacco products (12)	-2.394	0.246	-0.0059	0.1436	3,748
Textiles (13)	-0.579	0.072	-0.0009	0.0555	18,451
Wearing apparel (14)	-1.157	0.126	-0.0029	0.0687	11,913
Leather (15)	-3.883	0.285	-0.0050	0.0414	5,530
Wood products (16)	-3.036	0.265	-0.0058	0.0434	5,008
Paper (17)	-0.624	0.067	-0.0007	0.0254	6,760
Printing (18)	1.335	-0.035	0.0008	0.1045	3,062
Coke & refined petroleum (19)	0.245	0.055	-0.0012	0.0751	3,270
Chemicals (20)	-1.326	0.134	-0.0027	0.0695	17,611
Pharmaceuticals (21)	2.055	-0.129	0.0038	0.1363	9,747
Rubber & plastics (22)	-3.507	0.288	-0.0061	0.0362	12,846
Non-metallic minerals (23)	-1.012	0.099	-0.0015	0.0560	22,017
Basic metals (24)	-3.279	0.269	-0.0056	0.0330	14,932
Fabricated metal (25)	-4.862	0.371	-0.0079	0.0549	12,889
Electronics & optical (26)	-7.180	0.551	-0.0122	0.0835	5,371
Electrical equipment (27)	-3.232	0.247	-0.0048	0.0481	14,223
Machinery n.e.c. (28)	-1.735	0.145	-0.0024	0.0585	15,904
Motor vehicles (29)	-3.190	0.249	-0.0050	0.0401	12,484
Other transport equipment (30)	-0.556	0.038	0.0004	0.0512	5,136
Furniture (31)	-2.335	0.201	-0.0042	0.0488	3,391
Other manufacturing (32)	-2.151	0.196	-0.0040	0.0704	6,959